



UITs

**UNIVERSITY OF INFORMATION
TECHNOLOGY AND SCIENCES**

Project Proposal

Project Title: Blockchain-Based Supply Chain for Rice
& Essential Goods

Submitted by-	Submitted to-
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Overview:

This project aims to implement blockchain technology in the supply chain of rice and other essential goods in Bangladesh to enhance transparency, reduce fraud, and ensure fair pricing for both consumers and farmers. By utilizing Hyperledger or Ethereum, this system will provide an immutable, decentralized ledger that ensures traceability and efficiency in food distribution.

Key Terms:

Blockchain: A decentralized ledger that securely records transactions.

Smart Contracts: Self-executing contracts for automating supply chain agreements.

IoT Integration: Real-time tracking of goods using Internet of Things devices.

Supply Chain Transparency: End-to-end visibility of goods movement, reducing price manipulation.

Problem Statement:

Main Problem: The current food supply chain in Bangladesh faces inefficiencies, price manipulation, and a lack of transparency. Middlemen exploit both farmers and consumers, leading to artificial price inflation and supply disruptions.

Middlemen Exploitation: Unfair pricing practices affect farmers and consumers.

Lack of Traceability: No real-time tracking of food from farm to consumer.

Supply Chain Inefficiency: Manual processes cause delays and inaccuracies.

Adoption Barriers: Resistance from farmers and retailers towards digital solutions.

Uniqueness of the Solution:

Unlike traditional supply chains, this blockchain-based system will ensure transparency and fairness by eliminating unnecessary middlemen and providing real-time traceability of essential goods. Smart contracts will automate transactions, ensuring fair pricing and reducing delays.

Objectives:

1. Develop a blockchain-powered supply chain for essential goods.
2. Ensure fair pricing and reduced manipulation by middlemen.
3. Provide real-time tracking and verification of goods.
4. Improve efficiency and scalability in food distribution networks.

Proposed Solution:

Blockchain-Based Transparency: Immutable ledger ensuring trust.

IoT for Real-Time Tracking: Sensors for monitoring product movement.

Smart Contracts for Automation: Ensures timely and fair transactions.

User-Friendly Mobile Interface: Farmers, wholesalers, and consumers can access records easily.

Project Methodology:

Technology Stack:

Blockchain Platform: Hyperledger / Ethereum (Smart Contracts)

Front-End: React.js / Next.js

Back-End: Node.js / Laravel API

Storage: IPFS (for supply chain data security)

Development Timeline:

1. Weeks 1-2: Research & Requirements Gathering
2. Weeks 3-5: Blockchain and Smart Contract Development
3. Weeks 6-7: IoT Integration & Front-End Development
4. Week 8: System Integration & Testing
5. Week 9: Debugging & Performance Optimization
6. Weeks 10-11: Deployment, User Testing & Documentation

Resources:

Development Tools: Solidity (Ethereum), Hyperledger, Laravel, Next.js

Hosting: Decentralized storage (IPFS) & Cloud deployment

Database: Blockchain Ledger & IoT-enabled tracking system

Evaluation:

Performance Metrics: Track efficiency of transactions and supply chain transparency.

User Feedback: Collect feedback from farmers, retailers, and consumers.

Security Audits: Ensure blockchain integrity and transaction security.

Challenges & Limitations:

1. Farmer & Retailer Adoption: Training required for digital literacy.
2. Scalability Issues: Handling large transaction volumes on blockchain.
3. Government Integration: Collaboration needed with existing supply chain policies.

Sustainability Plan:

Regular Updates: Continuous improvement and system optimization.

Stakeholder Training: Educating farmers and retailers on blockchain benefits.

Backup & Recovery: Ensuring secure and resilient data storage.

Conclusion:

By integrating blockchain technology into Bangladesh's supply chain for rice and essential goods, this project will create a transparent, secure, and efficient food distribution system. The elimination of middlemen, real-time tracking, and automation will lead to fairer pricing, reduced fraud, and better food security for consumers and farmers alike.